Governing relations of the $CaCO_3$-barrier screen: derivation

Session P0-W3.1 (symbolic; no values)

Each section opens with the physical argument, and the algebra follows it. Every displayed relation carries its SI units in brackets. The notation is that of `theory/symbols.yaml`, the functions are those of `theory/contract.yaml`, and open choices for the sign audit are listed in `theory/open_questions.md` (cited here as Q1, Q2, ...).

1 Regimes and what each link must achieve

Physics. The proposed mechanism is assembled in one chemical environment and must survive in another. The template is anchored, binds Ca^{2+} and templates the mineral while the electrode is a water-processed, alkaline slurry at room temperature (R-slurry). The finished interface then sits in a non-aqueous, fluorine-free electrolyte with the Al collector held at the cathode's anodic potential (R-cell). The two regimes have different solvents, different reference states and different corrosion modes, so a relation is meaningful only in the regime whose reference states it uses. Every relation below is therefore tagged with its regime, and none is carried across without saying so.

Table 1 states, for each link, the necessary condition the link must meet in its regime. These are directions of inequalities, not thresholds: every numeric margin is registered at G2. Meeting them makes the mechanism thermodynamically possible under the stated framing; it does not show that a barrier works, and no rate, capacity or device statement follows from any of them.

Link	Regime	What the link must achieve	Relation
C1 anchor	R-slurry	template held at the oxide against water	$W_{\mathrm{adh,wet}} > 0$, Eq. (1)
C2 sequester	R-slurry	Ca^{2+} enriched at the layer against hydrated Ca^{2+}	$\Delta G_{\mathrm{seq}} < 0$, $\mathcal{E} > 1$, Eq. (7)
C3 nucleate	R-slurry	heterogeneous barrier lowered, and accessible at Ω_{local}	$f(\theta)$ small; $\Delta G^{*}_{\mathrm{het}}(\Omega_{\mathrm{local}})$ low, Eq. (18)
C4 conformal	R-slurry, R-cell	a continuous film that closes below the thickness bound	$S_{\mathrm{spread}} \geq 0$ or closure below $d_{\max}$, Eqs. (24), (31)
C5 stable	R-slurry	mineral and oxide on the aqueous Pourbaix hull	$\Delta G_{\mathrm{pbx}} = 0$ (tolerance at G2), Eq. (26)
C5 stable	R-cell	phases inside their Li-referenced window	$E_{\mathrm{red}} < E_{\mathrm{Li}} < E_{\mathrm{ox}}$, Eq. (28)

Table 1: Necessary conditions per link. C1–C3 are defined only in R-slurry: their reference states (water-wetted surfaces, hydrated Ca^{2+}, an aqueous ion activity product) do not exist in the cell.

Two rules follow from the regime split. First, C1–C3 make no statement about the cell: anchor retention, Ca^{2+} binding and nucleation in the organic electrolyte are not computed. Second, C5 needs two different constructions (Section 9), because aqueous Pourbaix analysis assumes water as the solvent and the hydrogen electrode as the reference.

2 Adhesion in water

Physics. In the slurry the template never meets a dry oxide. The hydroxylated Al_2O_3/AlOOH surface is covered by hydrogen-bonded water, and the polymer is hydrated too. A catechol–oxide contact can only form by displacing that water, so the relevant comparison is between two states that actually compete in the slurry: the oxide and polymer in contact, and the oxide and polymer each wetted by water. The dry (vacuum) work of adhesion compares the contact with two bare surfaces, a state that never occurs in R-slurry, and is therefore only a comparator.

The reversible work per unit area to separate a contact between phases A and B inside a medium M is the Dupré work $W = \gamma_{AM} + \gamma_{BM} - \gamma_{AB}$: the two new A–M and B–M interfaces are created and the A–B interface is destroyed. With A the oxide, B the polymer and M water,

$$W_{\mathrm{adh,wet}} = \gamma_{\mathrm{ow}} + \gamma_{\mathrm{pw}} - \gamma_{\mathrm{op}} \quad [\mathrm{J\,m^{-2}}]. \tag{1}$$

Units: each term is an interfacial free energy, so $W_{\mathrm{adh,wet}}$ is in $\mathrm{J\,m^{-2}}$. Writing every in-water energy through the dry works of each surface with water, $\gamma_{\mathrm{ow}} = \gamma_{\mathrm{o}} + \gamma_{\mathrm{w}} - W_{\mathrm{ow}}$ and $\gamma_{\mathrm{pw}} = \gamma_{\mathrm{p}} + \gamma_{\mathrm{w}} - W_{\mathrm{pw}}$, gives

$$W_{\mathrm{adh,wet}} = W_{\mathrm{adh,dry}} + 2\gamma_{\mathrm{w}} - W_{\mathrm{ow}} - W_{\mathrm{pw}}, \qquad W_{\mathrm{adh,dry}} = \gamma_{\mathrm{o}} + \gamma_{\mathrm{p}} - \gamma_{\mathrm{op}} \quad [\mathrm{J\,m^{-2}}]. \tag{2}$$

Equation (2) is the water-competition statement in one line: a large dry adhesion is offset by the works W_{ow} and W_{pw} with which water already binds each surface. A hydrophilic, hydroxylated oxide and a hydrophilic polymer bind water strongly, so $W_{\mathrm{adh,wet}}$ can be small or negative even when $W_{\mathrm{adh,dry}}$ is large. The bare-oxide control has $W_{\mathrm{adh,wet}} = 0$ by construction and fixes the zero of the scale.

The interfacial energies of Eq. (1) are not computed separately. The potential of mean force (PMF) of a template fragment along a separation coordinate in explicit water gives the free-energy difference between the separated and bound states, ΔF_{PMF} (J per fragment), which already contains the displacement penalty and the entropy of the released water. Normalising by the contact area,

$$W_{\mathrm{adh,wet}} = \frac{\Delta F_{\mathrm{PMF}}}{A_{\mathrm{c}}} \quad [\mathrm{J\,m^{-2}}], \tag{3}$$

with units $\mathrm{J/m^2}$. Two conditions make Eq. (3) meaningful: the PMF must reach a plateau at large separation (both surfaces fully rewetted), and A_{c} must be defined once for all fragments, because the ranking across binders depends on it (open question Q5). C1 requires $W_{\mathrm{adh,wet}} > 0$ beyond its uncertainty on the populated oxide terminations.

3 The sequestration cycle and the enrichment factor

Physics. The template can raise the local Ca^{2+} concentration only if it holds Ca^{2+} more strongly than water does. The question is a competition between two ligand environments for the same ion: the hydration shell in bulk solution and the coordinating groups of the template. The free energy must therefore start from hydrated Ca^{2+} at the regime activity, include the cost of preparing the binding site at the regime pH, and end in a well-defined local state.

Cycle and standard states. Let the coordinating group be an acid L–H that binds Ca^{2+} only when deprotonated. The cycle is

$$\begin{aligned}
&\text{(i)} \quad \mathrm{L{-}H\ (layer)} \ \rightarrow\ \mathrm{L^-(layer) + H^+(aq)}, && \Delta G_{\mathrm{i}} = -RT\ln\alpha \quad [\mathrm{J\,mol^{-1}}] \\
&\text{(ii)} \quad \mathrm{Ca^{2+}(aq,\ bulk) + L^-} \ \rightarrow\ \mathrm{Ca^{2+}\cdot L^-(layer)}, && \Delta G_{\mathrm{ii}} = \Delta G_{\mathrm{site}} \quad [\mathrm{J\,mol^{-1}}]
\end{aligned}$$

with $\alpha = 1/\left(1 + 10^{\,\mathrm{p}K_{\mathrm{a}} - \mathrm{pH}}\right)$ (dimensionless). Step (i) is the free energy of finding the site deprotonated when the solution fixes the proton activity: α is the deprotonated fraction at the regime pH,

and $-RT\ln\alpha \geq 0$. Step (ii) is the transfer of one Ca^{2+} from bulk solution into the binding layer. Summing,

$$\Delta G_{\mathrm{seq}} = \Delta G_{\mathrm{site}} + RT\ln\bigl(1 + 10^{\,\mathrm{p}K_{\mathrm{a}}-\mathrm{pH}}\bigr) \quad [\mathrm{J\,mol^{-1}}]. \tag{4}$$

Units: RT is in $\mathrm{J\,mol^{-1}}$ and the logarithm is of a pure number. The correction vanishes as $\mathrm{pH} - \mathrm{p}K_{\mathrm{a}} \to \infty$, equals $RT\ln 2$ at $\mathrm{pH} = \mathrm{p}K_{\mathrm{a}}$, and grows as the site becomes protonated. Both $\mathrm{p}K_{\mathrm{a}}$ and the activity coefficients are taken at the regime ionic strength. This is why catechol, largely protonated near neutral pH, must be tested rather than assumed (design flag 4).

Step (ii) fixes the standard states. Let $w(\mathbf{r})$ be the PMF of one Ca^{2+} relative to bulk solution (J per ion, $w \to 0$ far from the template), computed with counter-ions and water explicit. In the dilute limit the equilibrium concentration is $c(\mathbf{r}) = c_{\mathrm{bulk}}\,\mathrm{e}^{-w(\mathbf{r})/k_{\mathrm{B}}T}$. Assigning to each site a volume V_{site} (the binding-layer volume per site) and defining

$$\Delta G_{\mathrm{site}} \equiv -RT\ln\Bigl[\frac{1}{V_{\mathrm{site}}}\int_{\mathrm{site}} \mathrm{e}^{-w(\mathbf{r})/k_{\mathrm{B}}T}\,\mathrm{d}^3r\Bigr] \quad [\mathrm{J\,mol^{-1}}] \tag{5}$$

(the bracket is a ratio of volumes, so it is a pure number) makes the transfer free energy a partition between two concentration scales: the bulk at the regime activity on the solution side, and the mean concentration in the site volume on the layer side. The conventional binding free energy to a 1 mol-per-volume standard state c° differs from it by the standard-state correction

$$\Delta G^{\circ}_{\mathrm{bind}} = \Delta G_{\mathrm{site}} - RT\ln\bigl(N_{\mathrm{A}}c^{\circ}V_{\mathrm{site}}\bigr) \quad [\mathrm{J\,mol^{-1}}], \tag{6}$$

where $N_{\mathrm{A}}c^{\circ}V_{\mathrm{site}}$ is dimensionless. A literature binding constant can therefore be compared with ΔG_{site} only after Eq. (6), and only with the same V_{site}. The choice of V_{site} shifts ΔG_{site} by $RT\ln(V'_{\mathrm{site}}/V_{\mathrm{site}})$, so it is fixed once (open question Q4).

Activity versus concentration. Equation (5) defines a concentration ratio. If w contains every interaction of the ion with its surroundings, the chemical potential of Ca^{2+} is uniform at equilibrium and the activity in the layer equals the bulk activity; the whole enrichment then sits in a local activity coefficient. The screen instead uses a *local-ideal* bookkeeping: the local Ca^{2+} activity is the local concentration times the bulk activity coefficient, and the binding free energy is carried explicitly by ΔG_{seq}. This is a definition, not an approximation, provided the binding free energy is not counted a second time elsewhere (Section 7).

Enrichment and its validity range. With the local-ideal bookkeeping and dilute, non-interacting sites,

$$\mathcal{E} \equiv \frac{c_{\mathrm{local}}}{c_{\mathrm{bulk}}} = \exp\Bigl(-\frac{\Delta G_{\mathrm{seq}}}{RT}\Bigr) \quad [1]. \tag{7}$$

$\Delta G_{\mathrm{seq}} < 0$ gives enrichment ($\mathcal{E} > 1$); $\Delta G_{\mathrm{seq}} > 0$ gives depletion: the template is then a worse ligand than water. Equation (7) holds when (a) Ca^{2+} is dilute in bulk, (b) the sites are far from saturation, (c) one ion binds per site and sites do not interact, and (d) the local and bulk activity coefficients are equal. Condition (b) is quantified next.

Saturation bound. The layer has Γ_{s} sites per unit area in a thickness δ_{s}, so its Ca^{2+} content cannot exceed $c_{\max} = \Gamma_{\mathrm{s}}/\delta_{\mathrm{s}}$ ($\mathrm{mol\,m^{-2}}$ over m gives $\mathrm{mol\,m^{-3}}$). For independent, equivalent sites (Langmuir), the occupied fraction is $\Theta = x/(1+x)$ with $x = \mathcal{E}c_{\mathrm{bulk}}/c_{\max}$, and $c_{\mathrm{local}} = c_{\max}\Theta$:

$$c_{\mathrm{local}} = \frac{\mathcal{E}c_{\mathrm{bulk}}}{1 + \mathcal{E}c_{\mathrm{bulk}}/c_{\max}} \leq \min\bigl(\mathcal{E}c_{\mathrm{bulk}},\ c_{\max}\bigr) \quad [\mathrm{mol\,m^{-3}}]. \tag{8}$$

The dilute result (7) is the limit $x \ll 1$; at $x \gg 1$ the enrichment saturates at $c_{\max}/c_{\mathrm{bulk}}$ however negative ΔG_{seq} is. A strongly binding but sparse template is therefore bounded by its site density, not by its binding free energy. Electrostatic coupling between charged sites, which Langmuir ignores, is open question Q3.

4 Local supersaturation

Physics. Crystallisation is driven by the excess chemical potential of the dissolved ions over the solid. For a 1:1 salt that excess is set by the product of the two ion activities relative to its value at saturation. The template can change that product only where the nucleus forms, and only through the ion it binds.

One definition is used throughout:

$$\Omega = \frac{\mathrm{IAP}}{K_{\mathrm{sp}}} = \frac{a_{\mathrm{Ca}^{2+}}\, a_{\mathrm{CO}_3^{2-}}}{K_{\mathrm{sp}}} \quad [1], \qquad \Delta\mu = k_{\mathrm{B}}T \ln \Omega \quad [\mathrm{J}], \tag{9}$$

with K_{sp} the solubility product of the registered framing (ACC or calcite) and $\Delta\mu$ the chemical-potential drop per formula unit on crystallisation. Units: activities and K_{sp} are pure numbers on the molal scale; $k_{\mathrm{B}}T$ is in J. Nucleation requires $\Omega > 1$. The per-ion form $S = \Omega^{1/2}$, $\Delta\mu = 2k_{\mathrm{B}}T \ln S$, is the same statement and is not used, so that no factor of two can be lost between lanes. The charter's S for C3 is this Ω.

Enrichment of Ca^{2+} alone. In the local-ideal bookkeeping the local Ca^{2+} activity is $a_{\mathrm{Ca}^{2+}}\, c_{\mathrm{local}}/c_{\mathrm{bulk}}$, while the carbonate activity is held at its bulk value (carbonate is not assumed enriched). Then

$$\Omega_{\mathrm{local}} = \frac{c_{\mathrm{local}}}{c_{\mathrm{bulk}}}\,\Omega_{\mathrm{bulk}} \xrightarrow{\mathcal{E}\, c_{\mathrm{bulk}} \ll c_{\mathrm{max}}} \mathcal{E}\,\Omega_{\mathrm{bulk}} \quad [1], \qquad \Delta\mu_{\mathrm{local}} = \Delta\mu_{\mathrm{bulk}} - \frac{\Delta G_{\mathrm{seq}}}{N_{\mathrm{A}}} \quad [\mathrm{J}]. \tag{10}$$

Units: $\Delta G_{\mathrm{seq}}/N_{\mathrm{A}}$ is J per ion, matching $\Delta\mu$ per formula unit, since one Ca^{2+} enters each formula unit. The saturation bound carries over: $\Omega_{\mathrm{local}} \le (c_{\mathrm{max}}/c_{\mathrm{bulk}})\,\Omega_{\mathrm{bulk}}$.

Two cautions belong to this step. (1) A template that attracts Ca^{2+} is usually anionic and repels CO_3^{2-}. Holding $a_{\mathrm{CO}_3^{2-}}$ at its bulk value is then an upper estimate of Ω_{local}, biased in favour of the mechanism; the bound is acceptable for a screen only if it is reported as such (Q2). (2) Because the true chemical potential is uniform at equilibrium, the increase of Ω from Ω_{bulk} to Ω_{local} is the binding free energy moved into the driving force by the bookkeeping. It is legitimate only if the same free energy is not also credited through the interfacial energies (Section 7).

5 Homogeneous and heterogeneous nucleation

Physics. A cluster of the new phase gains $\Delta\mu$ for every formula unit it contains but pays γ for every unit of interface it creates. Volume grows as r^3 and surface as r^2, so small clusters are unstable and large ones grow; the barrier is the free energy at the crossover. On a substrate the cluster can replace part of its interface with a cheaper substrate contact, which lowers the barrier without changing the driving force.

Homogeneous barrier. In the capillarity approximation, a spherical cluster of radius r with molecular volume $v_{\mathrm{m}} = V_{\mathrm{m}}/N_{\mathrm{A}}$ has

$$\Delta G_{\mathrm{hom}}(r) = -\frac{4\pi r^3}{3}\frac{\Delta\mu}{v_{\mathrm{m}}} + 4\pi r^2 \gamma_{\mathrm{nl}} \quad [\mathrm{J}]. \tag{11}$$

Units: r^3/v_{m} counts formula units, times J per formula unit; $r^2\gamma_{\mathrm{nl}}$ is m^2 times $\mathrm{J\,m^{-2}}$. Setting $\partial\Delta G_{\mathrm{hom}}/\partial r = 0$,

$$r^* = \frac{2\gamma_{\mathrm{nl}} v_{\mathrm{m}}}{\Delta\mu} \quad [\mathrm{m}], \qquad \Delta G^*_{\mathrm{hom}} = \frac{16\pi\gamma_{\mathrm{nl}}^3 v_{\mathrm{m}}^2}{3\Delta\mu^2} = \frac{4\pi r^{*2}\gamma_{\mathrm{nl}}}{3} \quad [\mathrm{J}]. \tag{12}$$

Units: $\mathrm{J\,m^{-2}}\cdot\mathrm{m^3/J} = \mathrm{m}$ for r^*; $\mathrm{J^3\,m^{-6}}\cdot\mathrm{m^6/J^2} = \mathrm{J}$ for $\Delta G^*_{\mathrm{hom}}$. With $\Delta\mu = k_{\mathrm{B}}T\ln\Omega$, $\Delta G^*_{\mathrm{hom}} \to \infty$ as $\Omega \to 1^+$ (no driving force, no nucleation) and $\Delta G^*_{\mathrm{hom}} \to 0$ as $\Omega \to \infty$; it increases with γ_{nl} and V_{m} and decreases with T at fixed Ω. The absolute barrier in units of $k_{\mathrm{B}}T$ is $\Delta G^*_{\mathrm{hom}}/k_{\mathrm{B}}T$.

Spherical-cap geometry. On a flat substrate the cluster is a cap cut from a sphere of radius r, meeting the substrate at contact angle θ measured through the cap. Elementary geometry gives base radius $\rho = r\sin\theta$, height $h = r(1-\cos\theta)$, and

$$A_{\mathrm{cap}} = 2\pi r^2(1-\cos\theta), \qquad A_{\mathrm{base}} = \pi r^2\sin^2\theta \quad [\mathrm{m}^2], \tag{13}$$

$$V_{\mathrm{cap}} = \frac{\pi r^3}{3}(1-\cos\theta)^2(2+\cos\theta) \quad [\mathrm{m}^3]. \tag{14}$$

The cap's free energy has three parts: the volume term, the free nucleus–liquid surface, and the base, where substrate–liquid interface (γ_{sl}) is replaced by substrate–nucleus interface (γ_{sn}):

$$\Delta G_{\mathrm{het}}(r) = -V_{\mathrm{cap}}\frac{\Delta\mu}{v_{\mathrm{m}}} + \gamma_{\mathrm{nl}}A_{\mathrm{cap}} + (\gamma_{\mathrm{sn}}-\gamma_{\mathrm{sl}})A_{\mathrm{base}} \quad [\mathrm{J}]. \tag{15}$$

Stationarity of Eq. (15) with respect to θ at fixed cap volume is the contact-line balance, Young's relation $\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}} = \gamma_{\mathrm{nl}}\cos\theta$ (Section 6). Substituting it, the surface terms become

$$\gamma_{\mathrm{nl}}\pi r^2\big[2(1-\cos\theta) - \cos\theta\sin^2\theta\big] = \gamma_{\mathrm{nl}}\pi r^2(1-\cos\theta)^2(2+\cos\theta) = 4\pi r^2\gamma_{\mathrm{nl}}\,f(\theta) \quad [\mathrm{J}], \tag{16}$$

using $2-\cos\theta-\cos^2\theta = (1-\cos\theta)(2+\cos\theta)$, and the volume term is $\frac{4\pi r^3}{3}f(\theta)$ times $-\Delta\mu/v_{\mathrm{m}}$, with

$$f(\theta) = \frac{(2+\cos\theta)(1-\cos\theta)^2}{4} = \frac{2-3\cos\theta+\cos^3\theta}{4} \quad [1]. \tag{17}$$

Hence $\Delta G_{\mathrm{het}}(r) = f(\theta)\,\Delta G_{\mathrm{hom}}(r)$ at every r: the cap has the same critical radius as the sphere, and

$$\Delta G^*_{\mathrm{het}} = f(\theta)\,\Delta G^*_{\mathrm{hom}} \quad [\mathrm{J}]. \tag{18}$$

The factorisation holds only because one and the same γ_{nl} multiplies A_{cap} and enters Young's relation; this is the reason Section 6 keeps the bare γ_{nl} in Young.

Properties of f. $f(0) = 0$ (complete wetting, no barrier), $f(\pi/2) = \frac{1}{2}$ (hemisphere), $f(\pi) = 1$ (no wetting, the homogeneous barrier). Its derivative is $f'(\theta) = \frac{3}{4}\sin^3\theta \geq 0$ on $[0,\pi]$, so f is monotone and $0 \leq \Delta G^*_{\mathrm{het}} \leq \Delta G^*_{\mathrm{hom}}$ for every admissible θ. f is also the volume fraction of the cap relative to the full sphere of the same radius.

Assumptions. Capillarity (macroscopic interfacial energies at nucleus size), an isotropic γ_{nl} (closer to the amorphous framing than to faceted calcite), a flat, homogeneous substrate, no line tension, and a cluster of the bulk phase's composition. A non-classical pathway (prenucleation clusters, liquid precursors) is outside this framework and is a limitation, not a parameter.

6 Young's relation and the fixed sign convention

Physics. At the contact line, moving the line outward by a small amount replaces substrate–liquid interface by substrate–nucleus interface and stretches the nucleus–liquid surface. At equilibrium these balance. The contact angle therefore measures how much the substrate prefers the nucleus to the liquid, in units of the nucleus–liquid energy.

Convention, in words. θ is the angle at the three-phase line measured through the nucleus. $\cos\theta$ is the free energy gained per unit contact area by replacing substrate–liquid interface with substrate–nucleus interface, divided by the nucleus–liquid interfacial free energy. $\theta = 0$ means the nucleus wets the template completely; $\theta = \pi$ means it does not wet it at all.

Convention, as an equation.

$$\cos\theta = \frac{\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}}}{\gamma_{\mathrm{nl}}}, \qquad \gamma_{\mathrm{sn}} = \gamma_{\mathrm{sn}}^{(0)} + \Delta\gamma_{\mathrm{w}} \quad [1;\ \mathrm{J\,m^{-2}}]. \tag{19}$$

Units: a ratio of interfacial free energies. A finite angle exists only for $|\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}}| \le \gamma_{\mathrm{nl}}$. For $\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}} \ge \gamma_{\mathrm{nl}}$ the nucleus phase spreads ($S_{\mathrm{spread}} \ge 0$, Section 8) and $f = 0$; for $\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}} \le -\gamma_{\mathrm{nl}}$ it does not wet and $f = 1$. An implementation clips $\cos\theta$ to $[-1, 1]$ and reports which branch applied.

The water-displacement term. The template is hydrated. When the nucleus forms on it, the template-bound water layer under the contact disc must be displaced, and that costs free energy over the contact area only. Define

$$\Delta\gamma_{\mathrm{w}} \equiv \gamma_{\mathrm{sn}} - \gamma_{\mathrm{sn}}^{(0)} \quad [\mathrm{J\,m^{-2}}], \tag{20}$$

where $\gamma_{\mathrm{sn}}^{(0)}$ is a substrate–nucleus energy from a reference that does not include the removal of that layer (for example, a contact computed without the template's structured water). The sign convention is: $\Delta\gamma_{\mathrm{w}} > 0$ **is a cost.** From Eq. (19), $\partial\cos\theta/\partial\Delta\gamma_{\mathrm{w}} = -1/\gamma_{\mathrm{nl}} < 0$, and with $\partial f/\partial\cos\theta = -\frac{3}{4}\sin^2\theta \le 0$,

$$\frac{\partial f}{\partial\Delta\gamma_{\mathrm{w}}} = \frac{3\sin^2\theta}{4\gamma_{\mathrm{nl}}} \ge 0 \quad [\mathrm{m^2\,J^{-1}}]: \tag{21}$$

a positive displacement cost raises θ, f and $\Delta G^*_{\mathrm{het}}$, and $\Delta\gamma_{\mathrm{w}} < 0$ lowers them. If γ_{sn} is supplied directly as a full in-water interfacial free energy, the displacement is already inside it, $\Delta\gamma_{\mathrm{w}} \equiv 0$, and the term must not be added again. Applying it twice is the double-counting error that C3 item 13 demonstrates.

Why Young uses the bare γ_{nl} and not γ_{eff}. The deck writes $\gamma_{\mathrm{eff}} = \gamma_{\mathrm{nl}} - \Delta\gamma_{\mathrm{w}}$ and places it in the denominator of Young. That is rejected, for three reasons. (1) Geometry: the displaced water sits on the contact disc. The free surface of the cap, whose energy is $\gamma_{\mathrm{nl}}A_{\mathrm{cap}}$, never touches the template, so nothing about the template's hydration changes γ_{nl}. (2) Consistency: the factorisation of Eq. (18) requires the γ_{nl} in Young to equal the γ_{nl} that multiplies A_{cap} and sets $\Delta G^*_{\mathrm{hom}}$. A modified denominator would leave a residual surface term and break $\Delta G^*_{\mathrm{het}} = f\Delta G^*_{\mathrm{hom}}$. (3) Sign: with γ_{eff} in the denominator,

$$\left.\frac{\partial\cos\theta}{\partial\Delta\gamma_{\mathrm{w}}}\right|_{\mathrm{deck}} = \frac{\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}}^{(0)}}{(\gamma_{\mathrm{nl}} - \Delta\gamma_{\mathrm{w}})^2} \quad [\mathrm{m^2\,J^{-1}}], \tag{22}$$

which is positive on the wetting side ($\gamma_{\mathrm{sl}} > \gamma_{\mathrm{sn}}^{(0)}$): a displacement *cost* would lower the barrier. This is design flag 3. γ_{eff} is kept only as a named diagnostic, so that W4 can reproduce the deck's form for the double-counting demonstration. The contract encodes the reversal as sign-flip test SF1 (Table 2).

Change	Registered convention	Deck form (γ_{eff} in the denominator), $\gamma_{\mathrm{sl}} > \gamma_{\mathrm{sn}}^{(0)}$
$\Delta\gamma_{\mathrm{w}} > 0$	$\cos\theta\downarrow$, $\Delta G^*_{\mathrm{het}} \ge \Delta G^*_{\mathrm{het}}\vert_{\Delta\gamma_{\mathrm{w}}=0}$	$\cos\theta\uparrow$, $\Delta G^*_{\mathrm{het}} \le \Delta G^*_{\mathrm{het}}\vert_{\Delta\gamma_{\mathrm{w}}=0}$
$\Delta\gamma_{\mathrm{w}} < 0$	$\Delta G^*_{\mathrm{het}} \le \Delta G^*_{\mathrm{het}}\vert_{\Delta\gamma_{\mathrm{w}}=0}$	reversed
$\Delta G_{\mathrm{seq}} < 0$	$\Omega_{\mathrm{local}} > \Omega_{\mathrm{bulk}}$, barrier lowered	(not affected by the Young form)
$\Delta G_{\mathrm{seq}} > 0$	$\Omega_{\mathrm{local}} < \Omega_{\mathrm{bulk}}$, barrier raised	

Table 2: Sign-flip tests SF1 and SF2 of `theory/contract.yaml`.

7 Where the template acts, and the no-double-counting condition

Physics. The template is both an interface and a ligand field. As an interface it changes the energies of the surfaces the nucleus replaces and creates; as a ligand field it concentrates Ca^{2+}

where the nucleus forms. In classical nucleation the barrier depends on exactly four quantities, $(\gamma_{\mathrm{sl}}, \gamma_{\mathrm{sn}}, \gamma_{\mathrm{nl}}, \Delta\mu)$, so the template can act only through θ and through Ω. The risk is that one microscopic interaction, the binding of Ca^{2+} to the template, is counted in both.

Statement. Combining Eqs. (10), (12), (18) and (19),

$$\Delta G^{*}_{\mathrm{het}} = f\big(\theta[\gamma_{\mathrm{sl}}, \gamma^{(0)}_{\mathrm{sn}}, \Delta\gamma_{\mathrm{w}}]\big)\, \frac{16\pi\gamma^{3}_{\mathrm{nl}} v^{2}_{\mathrm{m}}}{3\big(k_{\mathrm{B}}T \ln \Omega_{\mathrm{local}}[\Delta G_{\mathrm{seq}}, \Gamma_{\mathrm{s}}, \delta_{\mathrm{s}}]\big)^{2}} \quad [\mathrm{J}]. \tag{23}$$

No-double-counting condition. (a) Every template-dependent input enters exactly one channel: $\gamma_{\mathrm{sl}}, \gamma^{(0)}_{\mathrm{sn}}, \Delta\gamma_{\mathrm{w}}$ only θ; $\Delta G_{\mathrm{seq}}, \Gamma_{\mathrm{s}}, \delta_{\mathrm{s}}$ only Ω; so $\partial\theta/\partial\Delta G_{\mathrm{seq}} = 0$ and $\partial\Omega/\partial\gamma_{\mathrm{s\cdot}} = 0$. Bulk inputs ($\gamma_{\mathrm{nl}}$, V_{m}, T) may enter both. (b) The interfacial energies γ_{sl} and γ_{sn} are evaluated for the template *without* the Ca^{2+} excess that Ω_{local} already credits. Part (a) is checked mechanically by `theory/tools/check_contract.py`; part (b) is the physical content and is proved in P0-W3.2.

Outline of the proof (P0-W3.2).

1. *Exact reference.* Write the free energy of a cap on the template in the grand ensemble at the bulk chemical potentials of Ca^{2+} and CO_3^{2-}. In this bookkeeping (A) the driving force is $\Delta\mu_{\mathrm{bulk}}$ everywhere, and all template–ion interactions are inside grand-canonical interfacial energies $\gamma^{(\mu)}_{\mathrm{sl}}$, $\gamma^{(\mu)}_{\mathrm{sn}}$.

2. *Gibbs adsorption.* At fixed T, $\mathrm{d}\gamma^{(\mu)}_{\mathrm{sl}} = -\sum_i \Gamma_i\, \mathrm{d}\mu_i$, so the Ca^{2+} excess Γ_{Ca} at the template lowers $\gamma^{(\mu)}_{\mathrm{sl}}$ relative to the Ca-free template by $\int \Gamma_{\mathrm{Ca}}\, \mathrm{d}\mu_{\mathrm{Ca}}$; the same holds at the substrate–nucleus interface with its own excess.

3. *Local-ideal bookkeeping (B), the one registered here.* Move the binding free energy out of the interfacial energies and into the driving force: $\Delta\mu_{\mathrm{local}}$ from Eq. (10), with $\gamma_{\mathrm{sl}}, \gamma_{\mathrm{sn}}$ of the Ca-free template.

4. *Equivalence or bound.* Show under which conditions (A) and (B) give the same $\Delta G^{*}_{\mathrm{het}}$, or bound their difference. The key quantity is the Ca^{2+} excess at the substrate–liquid interface *minus* that at the substrate–nucleus interface: in (A), adsorption that stabilises only the template–water interface *raises* θ; it lowers the barrier only if the nucleus incorporates the bound ions at a lower cost than water holds them.

5. *Error demonstration.* Mixing (A)'s Ca-loaded γ_{sl} with (B)'s Ω_{local} counts the same binding twice; the size of that error is C3 item 13, together with the $\Delta\gamma_{\mathrm{w}}$ double application of Section 6.

Step 4 matters for the screen, and is flagged as an open question (Q1), because bookkeeping (B) always lowers the barrier when $\Delta G_{\mathrm{seq}} < 0$, while in (A) the sign of the template's effect depends on a difference of adsorption excesses. The choice of bookkeeping is therefore not neutral, and the proof must settle it before any C3 output is read.

8 Film conformality

Physics. A barrier must be continuous: one through-path lets electrolyte reach Al. A low nucleation barrier says where nuclei form easily, not whether they spread into a film. Whether the mineral spreads or beads on the template is a wetting balance, and whether beads merge before they grow thick is a competition between how many islands form and how far each spreads.

Spreading coefficient. The free-energy change per unit area when a thick layer of the nucleus phase replaces the template–liquid interface is

$$S_{\mathrm{spread}} = \gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}} - \gamma_{\mathrm{nl}} = \gamma_{\mathrm{nl}}(\cos\theta - 1) \quad [\mathrm{J\,m^{-2}}], \tag{24}$$

the second form holding on the partial-wetting branch where Young has a solution. Units: a sum of interfacial free energies. Since $\cos\theta \le 1$, a finite contact angle always has $S_{\mathrm{spread}} \le 0$. The sign condition is: $S_{\mathrm{spread}} \ge 0$ means no finite angle exists, $\theta \to 0$, $f \to 0$, and the phase grows layer by layer (Frank–van der Merwe); $S_{\mathrm{spread}} < 0$ means $\theta > 0$ and growth by islands (Volmer–Weber). Island-then-layer growth (Stranski–Krastanov) needs a strain or misfit energy that this isotropic framework does not contain. The registered $\Delta\gamma_{\mathrm{w}}$ enters S_{spread} through γ_{sn}, with the same sign as in θ: a displacement cost makes spreading harder.

Coalescence of islands. With $S_{\mathrm{spread}} < 0$, a film forms only when islands meet. Let the islands be equal caps of contact angle θ on a square lattice of density N_{s} (spacing $\ell = N_{\mathrm{s}}^{-1/2}$). A cap of base radius ρ has height $h = \rho\tan(\theta/2)$. Neighbours touch when $2\rho = \ell$, and the base discs then cover only $\pi/4$ of the area; the square cells are covered when $\rho = \ell/\sqrt{2}$. The two geometric thicknesses are

$$h_{\mathrm{c}} = \frac{\tan(\theta/2)}{2\sqrt{N_{\mathrm{s}}}}, \qquad h_{\mathrm{cl}} = \frac{\tan(\theta/2)}{\sqrt{2N_{\mathrm{s}}}} = \sqrt{2}\,h_{\mathrm{c}} \quad [\mathrm{m}]. \tag{25}$$

Units: $N_{\mathrm{s}}^{-1/2}$ is a length. Both vanish as $\theta \to 0$ and diverge as $\theta \to \pi$. They are geometric estimates that treat caps as growing self-similarly up to contact; they are arguments, not measurements, and N_{s} comes from a nucleation-and-growth model (item 17), not from thermodynamics.

What a contact angle does and does not imply about coverage. A small θ means thin islands and early closure for a given N_{s}. It does not fix N_{s}, which depends on the nucleation rate relative to lateral growth; it says nothing about places where the template is absent, where the bare-oxide angle applies; and it is silent on porosity, on dissolution and reprecipitation of the amorphous phase, and on cracking when a hydrated film shrinks on drying. The proposed C3 band $f(\theta) \le \frac{1}{2}$ admits θ up to $\pi/2$, which is island growth; conformality is therefore a separate criterion (C4, design flag 2), chosen at G2 among a small-angle bound, $S_{\mathrm{spread}} \ge 0$, and a closure condition such as Eq. (31).

9 Thermodynamic stability

Physics. A layer that is thermodynamically unstable in its own environment dissolves or reacts. The environment differs between regimes, so stability is two questions: in the slurry, whether $CaCO_3$ and the native oxide survive alkaline water at the regime pH and potential; in the cell, whether the phases survive the Li reservoir at the collector's potential. Both are screens: necessary, not sufficient, and blind to morphology and kinetics.

R-slurry: aqueous Pourbaix. Each species ϕ (solid or dissolved) is brought to a common reference by exchanging O and H with water, protons and electrons at the regime pH and potential. Per non-O, non-H atom,

$$\Psi_\phi = \frac{1}{n_{\mathrm{M}}}\Big[G_\phi - n_{\mathrm{O}}\mu_{\mathrm{H_2O}} - RT\ln 10\,(2n_{\mathrm{O}} - n_{\mathrm{H}})\,\mathrm{pH} \\ - (2n_{\mathrm{O}} - n_{\mathrm{H}} + Q)FE_{\mathrm{SHE}}\Big] \quad [\mathrm{J\,mol^{-1}}], \qquad \Delta G_{\mathrm{pbx}} = \Psi_\phi - \Psi_{\mathrm{hull}} \ge 0. \tag{26}$$

Units: G_ϕ and $\mu_{\mathrm{H_2O}}$ are $\mathrm{J\,mol^{-1}}$, $RT\ln 10\,\mathrm{pH}$ is $\mathrm{J\,mol^{-1}}$, and FE_{SHE} is $\mathrm{C\,mol^{-1}{\cdot}V} = \mathrm{J\,mol^{-1}}$. Species with $2n_{\mathrm{O}} - n_{\mathrm{H}} + Q > 0$ are favoured by raising E_{SHE}, and species with $2n_{\mathrm{O}} - n_{\mathrm{H}} > 0$ by

raising pH. Ψ_{hull} is the lower envelope over all species and mixtures of the same metal composition, and $\Delta G_{\mathrm{pbx}} = 0$ means stable. *What it can say:* whether the phase is the equilibrium state at the chosen (pH, E_{SHE}) and dissolved-ion activities. *What it cannot say:* rates, passivation, local pH at the interface, morphology. Two traps: the amorphous framing lies above the calcite hull by its offset, so ACC is never on the hull and the tolerance must be registered (C5 open item); and the dissolved-ion activities chosen for the aqueous species decide whether solids appear stable at all, so the screen can be satisfied by construction unless it is shown to fail where it must (C5 item 21, Q8).

R-cell: Li-open grand potential. In the cell the solvent is not water and the reference is $\mathrm{Li}^+/\mathrm{Li}$. Holding the collector at E_{Li} fixes the Li chemical potential,

$$\mu_{\mathrm{Li}} = \mu^{\circ}_{\mathrm{Li}} - F E_{\mathrm{Li}} \quad [\mathrm{J\,mol^{-1}}], \qquad \Phi_{\phi} = G_{\phi} - n_{\mathrm{Li}} \mu_{\mathrm{Li}} \quad [\mathrm{J\,mol^{-1}}]. \tag{27}$$

For a reaction $\phi \to \sum_k \nu_k \psi_k + \Delta n\,\mathrm{Li}$ that releases Δn Li to the reservoir, Li conservation gives $\Delta\Phi(E_{\mathrm{Li}}) = \sum_k \nu_k \Phi_k - \Phi_{\phi} = \Delta G_{\mathrm{r}} - \Delta n\, F E_{\mathrm{Li}}$, with ΔG_{r} the reaction free energy against Li metal. The phase is stable against that reaction while $\Delta\Phi > 0$, so each reaction sets one limit,

$$E^{*} = \frac{\Delta G_{\mathrm{r}}}{\Delta n\, F} \quad [\mathrm{V}], \qquad \begin{cases} \Delta n > 0 \text{ (delithiation, oxidation)}: & E_{\mathrm{Li}} < E^{*} \;\to\; E_{\mathrm{ox}} = \min E^{*}, \\ \Delta n < 0 \text{ (lithiation, reduction)}: & E_{\mathrm{Li}} > E^{*} \;\to\; E_{\mathrm{red}} = \max E^{*}. \end{cases} \tag{28}$$

Units: $\mathrm{J\,mol^{-1}}$ over $\mathrm{C\,mol^{-1}}$ is V. *What it can say:* whether a phase reacts with Li at the collector's potential. *What it cannot say:* oxidation that exchanges no Li. A Li-free phase such as $CaCO_3$ has no reaction with $\Delta n > 0$, so its Li-open E_{ox} is unbounded: the construction declares it stable at any anodic potential by construction. Anodic stability of $CaCO_3$ and of the Al phases needs oxidation reactions that release CO_2, O_2 or dissolved Ca^{2+} and Al^{3+} into the electrolyte, which require the electrolyte species as additional open components (Q9). Kinetics and electrolyte oxidation are outside both screens.

10 The insulating-film limitation

Physics. $CaCO_3$ is an electronic insulator. The film lies between the cathode coating and the Al collector, and every electron of the cathode current must cross it. A film that is continuous but too thick blocks the collector, so conformality cannot be bought with thickness. Two transport limits bound the admissible thickness; the numbers come later, through macros.

Ohmic limit. If electrons cross by bulk conduction with resistivity ρ_{e}, the area-specific resistance and the bound set by an admissible $R_{A,\max}$ are

$$R_A = \rho_{\mathrm{e}} d \quad [\Omega\,\mathrm{m^2}], \qquad d \le d^{\mathrm{ohm}}_{\max} = \frac{R_{A,\max}}{\rho_{\mathrm{e}}} \quad [\mathrm{m}]. \tag{29}$$

Units: $\Omega\,\mathrm{m}$ times m is $\Omega\,\mathrm{m^2}$; $\Omega\,\mathrm{m^2}$ over $\Omega\,\mathrm{m}$ is m.

Tunnelling limit. For a thin film the current is carried by direct tunnelling through a barrier of height ϕ_{B}. In the WKB approximation for a rectangular barrier the transmission decays as $\mathrm{e}^{-2\kappa d}$, so

$$j = j_0\, \mathrm{e}^{-2\kappa d} \quad [\mathrm{A\,m^{-2}}], \qquad \kappa = \frac{\sqrt{2 m_{\mathrm{e}} \phi_{\mathrm{B}}}}{\hbar} \quad [\mathrm{m^{-1}}], \qquad d \le d^{\mathrm{tun}}_{\max} = \frac{\ln(j_0/j_{\mathrm{req}})}{2\kappa} \quad [\mathrm{m}]. \tag{30}$$

Units: $\sqrt{\mathrm{kg \cdot J}}/(\mathrm{J\,s}) = \mathrm{m^{-1}}$; the logarithm is of a ratio of current densities. The bound is positive only for $j_0 > j_{\mathrm{req}}$. The prefactor j_0, the effective mass and the barrier height are inputs with large uncertainty (Q10), and defects or hopping can only raise the current, so Eq. (30) is a conservative bound for a defect-free film.

Combined with conformality. The film must close before it exceeds the admissible thickness:

$$h_{\mathrm{cl}} = \frac{\tan(\theta/2)}{\sqrt{2N_{\mathrm{s}}}} \leq d_{\max} = \max\left(d_{\max}^{\mathrm{ohm}},\, d_{\max}^{\mathrm{tun}}\right) \quad [\mathrm{m}]. \tag{31}$$

This is a stated limitation (C4 item 18), not a performance figure. It contains no statement about cell resistance, rate or capacity.

11 Summary: links and the inputs they consume

Physics. Each link reduces to a small set of inputs. Listing them once is what lets the parameter table (W1) fetch every input and the implementation (W4) consume every one, with nothing typed and nothing missing. Table 3 is that list; the ids are those of `theory/symbols.yaml`, and the functions are those of `theory/contract.yaml`.

Link	Regime	Contract functions	Inputs consumed (symbol ids)	Source
C1 adhesion	R-slurry	`W_adh_wet`, `W_adh_wet_from_pmf`, `W_adh_dry_fn`	`Delta_F_pmf`, `A_c`; comparator `gamma_o`, `gamma_p`, `gamma_op`, `gamma_w`	computed PMF; W1 for comparators
C2 sequestration	R-slurry	`Delta_G_seq_cycle`, `E_enr`, `c_max`, `c_local`	`Delta_G_site`, `pKa`, `pH`, `T`, `c_bulk`, `Gamma_s`, `delta_s`, `V_site`	computed PMF; W1 for pKa, pH, c_bulk
Super-saturation	R-slurry	`Omega_bulk`, `Omega_local`, `Delta_mu`	`a_Ca`, `a_CO3`, `K_sp`, `T`, with C2 outputs	W1 (activities from speciation at the regime pH and ionic strength)
C3 nucleation	R-slurry	`r_star`, `DG_hom`, `f_theta`, `DG_het`, `cos_theta_young`, `theta_young`; audit: `gamma_eff_deck`, `cos_theta_deck`	`gamma_nl`, `gamma_sl`, `gamma_sn0`, `Delta_gamma_w`, `V_m`, `T`, with Ω_{local}	W1 priors and computed slabs; framing at G2
C4 conformality	R-slurry	`S_spread`, `h_c`, `h_cl`	C3 energies, `N_s`	nucleation-and-growth model (item 17)
C4 film bound	R-cell	`R_A`, `d_max_ohm`, `kappa`, `d_max_tun`	`rho_e`, `R_A_max`, `phi_B`, `m_e`, `j_0`, `j_req`, `d_film`	W1; `R_A_max`, `j_req` registered at G2
C5 Pourbaix	R-slurry	`Psi_pbx`, `Delta_G_pbx`	`G_phase` (with the ACC offset), `mu_H2O`, `N_O`, `N_H`, `N_M`, `Q_charge`, `pH`, `E_SHE`, `T`	W1 thermochemistry; materials database
C5 window	R-cell	`mu_Li`, `Phi_Li`, `E_window`	`G_phase`, `N_Li`, `mu_Li0`, `E_Li`, `Delta_G_r`, `Delta_n_Li`	materials database

Table 3: Each link, the contract functions that implement it, and the inputs it consumes. Every input is fetched or computed; none is a result of the study.